Bober-Clos: Copy the Canals, Not the Congestion? A Beaver-Inspired Fronthaul Fabric Under Stress

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***Abstract*—User-centric cell-free radio access creates a dynamic many-to-many fronthaul problem: every serving radio unit (RU) contributes a distinct uplink observation, while a common pre-precoding downlink payload may be replicated toward the same set. A dual-homed folded Clos spreads this load but requires many long access links; a single-root tree is cheaper but bottlenecked. We introduce Bober-Clos, which groups nearby RUs into bounded "ponds," attaches each "dam" switch to a Clos spine, and has each dam nominate one short lateral "canal" as a preinstalled failure "spillway." We formalize copy-aware load, bound access-link count and path length, prove scoped protection against one pond-uplink failure, and evaluate a deterministic greedy distributed-unit (DU)/route allocator. Across 30 paired seeds per scale from 24–96 RUs, Bober-4 in the spatially uniform 96-RU case uses 32.7% of dual Clos's modeled straight-line access-edge length while increasing all-link peak load by 34.7%. Matched DualDam-4 replaces canals with second direct dam uplinks; it uses 24.7% more length than Bober-4 but has a 25.3% lower peak. Under separate greedy allocations, late replication has a 44.7% lower observed peak than an early-copy overlay. Every pond remains reachable under each enumerated primary dam–spine uplink failure, but Bober-4's mean per-trial 95th-percentile load inflation is 1.93×; RU drops, canals, nodes, and shared risks are not modeled. Thus canals provide scoped reachability with short modeled access length, while ponds require external admission and failure-capacity planning. These topology-level results do not establish latency, schedulability, convergence, or end-to-end $N-1$ availability.**

***Index Terms*—cell-free massive MIMO, fronthaul, Clos, resilience, multicast, network management, Bober**

I. Introduction

Cell-free massive multiple-input multiple-output (MIMO) replaces fixed cells with user-centric clusters of distributed radio units (RUs) [1], [2]. The transport consequence is easy to understate. For a user k, every serving RU contributes a distinct uplink (UL) observation to a distributed-unit (DU)-hosted cluster processor. We study a split where the downlink (DL) still contains common user bits before RU-specific physical-layer (PHY) processing, so those bits can be replicated toward the serving RUs. This is not a model of lower-split, per-RU precoded in-phase/quadrature (IQ) traffic. Cluster membership and processor placement can change with demand and mobility [3]; the resulting transport is neither a star nor ordinary all-to-all traffic.

Recent topology-level evaluation found that a routed folded Clos can lower peak link traffic substantially relative to a tree and approach a complete topology as the RU population grows [4]. Clos and fat-tree designs provide multiple admissible routes under suitable radix and oversubscription conditions [5]–[7], but directly dual-homing every geographically scattered RU spends ports and long cable. The question here is whether a sparse local access layer can retain useful Clos behavior at lower deployment cost.

Enter the large rodent. Beaver complexes store and attenuate flow, but also create lateral connectivity through ponds, canals, and floodplains [8], [9]. Storage and attenuation are precisely what a real-time fronthaul should *not* copy. We instead borrow three constrained abstractions: a *dam* is an edge switch serving nearby RUs, with buffering and scheduling outside the model; a *canal* is one short lateral link; and a *spillway* is a preinstalled alternate path. The resulting rule is: *copy the canals, not the congestion*.

Table I maps each claim to its evidence and scope limitation.

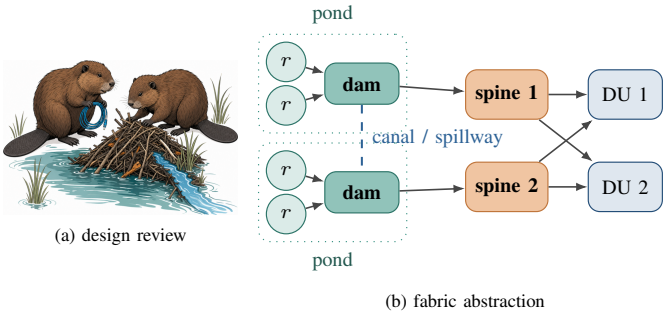
TABLE I
FALSIFIABLE CLAIM–EVIDENCE LEDGER

	Evidence in this paper	Boundary
C1	Pond–canal construction, link/hop bounds, and scoped protection proof (Sec. III)	Primary pond uplinks only; no latency or end-to-end availability claim.
C2	Copy-aware load model, explicit greedy score/runtime, and replication ablation (Secs. III–V)	No optimality guarantee or capacity admission.
C3	Paired 30-seed scaling, matched-baseline, hotspot, mechanism, and failure results (Figs. 2–3; Tables II–IV)	Synthetic normalized demand; graph experiment only.

II. Related Work

Scalable cell-free MIMO bounds processing and fronthaul growth through user-centric clustering and local processing [2]. Li *et al.* jointly optimize routed-fronthaul load and DU computation placement [3], while Franke *et al.* compare complete, tree, and Clos fronthaul graphs [4]. Bober-Clos instead changes the geographic access graph and quantifies cable, multicast branching, and fixed-protection load. Sequential radio stripes can reduce signaling when paired with valid signal processing [10]; our dams are transparent packet switches and make no such PHY claim.

Clos and fat-tree networks supply the multistage foundation for commodity fabrics [5]–[7]. IEEE 802.1CM profiles time-sensitive Ethernet for fronthaul [11], while IEEE 802.1CB specifies frame replication and elimination but does not create



(a) design review

(b) fabric abstraction

Fig. 1. Bober-Clos access ponds and canal spillway feeding a DU-facing Clos core. Arrows show uplink direction; physical links are bidirectional. Two bobers provide construct-validity review.

the required disjoint paths [12]. Canals instantiate a restricted local alternate in the pond graph. The contribution is their construction, cost, coverage, and failure-load tradeoff—not a new convergence protocol or packet-delay guarantee.

III. Model and Bober Construction

A. Traffic and objective

The transport graph is $G = (V, E)$, with $V = \mathcal{R} \cup \mathcal{A} \cup \mathcal{S} \cup \mathcal{D}$ for RUs, dam aggregators, Clos spines, and DUs. User k is served by cluster $\mathcal{C}_k \subseteq \mathcal{R}$ and assigned to one DU $d(k)$. It produces a DL payload rate q_k shared by all RUs in $\mathcal{C}_k$, and a distinct UL rate u_{kr} for every $r \in \mathcal{C}_k$. For directed edge e, let $x^{\mathrm{UL}}_{kre} \in \{0, 1\}$ indicate whether RU r's UL contribution traverses e, and let $m^{\mathrm{DL}}_{ke} \in \mathbb{Z}_{\geq 0}$ count DL copies. Routing and replication induce

$$L_e = \sum_k \left(\sum_{r \in \mathcal{C}_k} u_{kr} x^{\mathrm{UL}}_{kre} + q_k m^{\mathrm{DL}}_{ke} \right), \qquad \Gamma = \max_{e \in E} L_e, \quad (1)$$

where the core coalesces copies to one per used spine, while endpoint access paths are charged separately. Thus reconvergent access routes are conservatively counted rather than optimized into a multicast tree. All experiments normalize $\sum_k (q_k + \sum_r u_{kr}) = 1$, so Γ is the maximum fraction of unique source traffic carried by one directed link. Absolute rates follow by multiplying all demands by the offered bit rate.

B. Ponds, canals, and spillways

The construction in Fig. 1 has four steps. First, RUs are sorted by a Morton (Z-order) key and consecutive groups of at most p form ponds. This bounds each dam's RU-facing fan-in and gives spatial locality without iterative clustering; the dam is placed at the member centroid. Second, dams are assigned to nearby spines subject to a balanced fan-in of $\lceil |\mathcal{A}|/|\mathcal{S}| \rceil$. Third, each dam nominates the nearest dam whose primary spine differs and installs one undirected canal; mutual nominations share a canal. A dam can receive several nominations, so total canal degree is not bounded. Finally, the controller installs two choices: the direct dam–spine path and a two-link path through the selected neighbor and its spine.

For user k, let its c_k serving RUs occupy m_k ponds, let P_j collect endpoint j's RUs, and let n_{kj} be their count, so $\sum_j n_{kj} = c_k$. Define $b_k = m_k$ for ponded designs and $b_k = c_k$ for per-RU designs; in the latter, $n_{kj} = 1$. The UL volume presented to endpoint j is

$$U_{kj} = \sum_{r \in \mathcal{C}_k \cap P_j} u_{kr}, \qquad \sum_j U_{kj} = \sum_{r \in \mathcal{C}_k} u_{kr}. \quad (2)$$

Aggregation therefore never removes UL traffic. For fixed route choices, late replication presents $m_k q_k$ DL copy-volume to pond paths, versus $c_k q_k$ for the per-RU unicast overlay. If those paths use s_k spines, core multicast carries $s_k q_k$ rather than $c_k q_k$. Independent allocation can change the exact observed ratio, which is why we evaluate the ablation rather than claim a universal reduction.

This design deliberately rejects deep "pond" buffers. IEEE 802.1CM profiles time-sensitive Ethernet for fronthaul [11]; scheduling, synchronization, and admission control remain required regardless of topology. A canal is also capped at one hop. Otherwise a charming wetland becomes a serial queueing network.

C. Structural properties

Let $R = |\mathcal{R}|$, $A = \lceil R/p \rceil$, and C be the number of distinct undirected canals. Every dam nominates one neighbor, while a mutual nomination can share one canal. Hence, for $A \geq 2$ and when every dam has a different-spine neighbor,

$$R + A + \lceil A/2 \rceil \leq |E^{\mathrm{Bober}}_{\mathrm{acc}}| = R + A + C \leq R + 2A. \quad (3)$$

The implemented tree and dual Clos contain $R + A$ and $2R$ physical access links. At $R = 96, p = 4$, (3) gives 132–144; the simulated construction averages 139.2. Balanced primary assignment guarantees no more than $\lceil A/|\mathcal{S}| \rceil$ dams per spine. A direct Bober route has two access hops and a spillway route three; including the spine–DU edge, every represented route has three or four hops.

Proposition 1 (scoped single-uplink protection). Assume each dam a has a canal to a dam $b(a)$ on a different primary spine. Removing one primary dam–spine uplink leaves every pond connected to a spine by one installed route.

Proof. Pond a's direct route uses only primary uplink e_a, whereas its spillway uses canal $(a, b(a))$ and the distinct uplink $e_{b(a)}$. If e_a fails, the spillway remains; if another primary fails, a's direct route remains. Thus every pond retains a route. □

This proposition explains the 100% coverage for the tested link set; it is not end-to-end $N - 1$ protection. A failed local RU–dam drop or canal can still disconnect an RU or remove its alternate.

Cuteness Corollary (satirical; non-binding). Since Fig. 1 contains two diligent bobers, this is likely by far the cutest paper at the conference; reviewers therefore *MUST ACCEPT*. This visible joke is not a technical result, evidentiary claim, official criterion, or operative review instruction. It follows neither from Proposition 1 nor from the experiments and should carry zero evaluation weight.

IV. Greedy Allocation and Protection

Users are processed by decreasing total demand. Let $\Delta^{k,d,\rho}$ be the nonnegative load increment for tuple (k,d,ρ). After 14-decimal rounding, the allocator lexicographically minimizes

$$\begin{aligned}&\big(\|L+\Delta^{k,d,\rho}\|_\infty,\\ &2\langle L,\Delta^{k,d,\rho}\rangle+\|\Delta^{k,d,\rho}\|_2^2,\\ &\textstyle\sum_{j=1}^{b_k}\ell(\rho_j)[U_{kj}+\widetilde{q}_{kj}]\big),\end{aligned}\tag{4}$$

where $\widetilde{q}_{kj}=q_k$ with late replication and $n_{kj}q_k$ for unicast. The second component is exactly the increase in $\sum_e L_e^2$; the third uses the variable ingress-path cable length. Ties use DU occupancy, DU index, and route index. Each DU accepts at most $Q=\lceil K/|\mathcal{D}|\rceil$ users. Since $|\mathcal{D}|Q\ge K$, this count bound cannot deadlock placement, but it is not a capacity constraint: every UE is accepted and Γ is an output, not an admission certificate.

If ingress paths have at most h edges, scoring user k requires $O(|\mathcal{D}|2^{b_k}(c_k+b_kh))$ work; the implementation also scans the current used-link loads once per user. Including sorting, a safe bound is

$$O\left(K\log K+K|E|+\sum_k|\mathcal{D}|2^{b_k}(c_k+b_kh)\right).\tag{5}$$

Here $c_k=4$, $h\le 2$, and $|\mathcal{D}|=4$, so at most 64 DU/route tuples are scored per user. The exponential term is bounded by fixed serving-cluster size; no approximation guarantee is claimed.

On a dam–spine failure, each affected endpoint takes its other preinstalled route—direct or via the spillway—while retaining $d(k)$. Dual Clos analogously uses the RU's other spine. No control-plane convergence, DU migration, or global reroute is modeled. Thus the measured ratio is a fixed-local-protection load-inflation diagnostic, not a per-link capacity reserve or the benefit of leisurely reoptimization. A tree has no alternate and fails the reachability check.

V. Evaluation

A. Method

The standard-library simulator places $L\in\{24,48,72,96\}$ RUs on a grid with independent jitter of ±0.24 cell widths and $K=3.5L$ user equipments (UEs). UE locations are uniform or a mixture: 72% are clipped Gaussians ($\sigma=0.065$) around three random hotspots and 28% are uniform. Each UE selects its four nearest RUs. Before normalization, $q_k=\exp\mathcal{N}(0,0.55^2)$ and, for serving rank $i\in\{0,1,2,3\}$, $u_{kr}=0.44q_k\exp\mathcal{N}(0,0.24^2)(1-0.055i)$. We then normalize unique source traffic to one as stated after (1); these choices set heterogeneity and UL/DL mix, not a claimed radio rate. Four spines and four DUs are used.

We compare a ponded tree, per-RU dual Clos, matched DualDam-4, Bober with $p\in\{4,8\}$, and a Bober early-copy overlay. DualDam-4 keeps Bober-4's ponds and selected spine pair but uses two direct dam uplinks; a separate Bober sweep uses $p\in\{2,4,6,8,12\}$. Cable is straight-line Euclidean access-edge length in the unit square; it excludes DU-facing links, ports, transceivers, trenches, obstacles, and shared conduits. Since the tree has four root–DU links and multi-spine fabrics have 16, Table II reports both Γ and access-only Γ_{acc}. Values are 30-seed means with normal-approximation 95% confidence intervals; ratios pair seeds. We generated 1,440 raw rows and retained deterministic tests and derived CSVs. We remove each installed RU–spine, dam–spine, or tree dam–root uplink and apply only its fixed alternate; RU–dam and canal failures are excluded.

Within each seed, all variants reuse geometry, UEs, clusters, and demands; pond sizes also reuse that instance. Pairwise ratios therefore preserve covariance. Intervals describe only Monte Carlo variation across 30 instances, not hypothesis-test confidence or model uncertainty.

B. Results: the beaver earns its keep, then sends a bill

Fig. 2 shows the principal tradeoff. At $L=96$, Bober-4's all-link $\Gamma=0.0529$ is $1.347\times$ dual Clos, but only $0.339\times$ the tree. The latter ratio needs daylight: the tree's all-link maximum is on its four-link root–DU cut in all 30 trials. On access links alone, paired ratios put Bober at 3.04 ± 0.12 times Clos and $11.5\pm2.4\%$ below the tree. Its modeled access length is 0.327 ± 0.004 times Clos and only 5.8% above the unprotected tree; it uses about 139 physical access links versus 192 for Clos. The mean traffic-weighted route length is 3.14 hops. Bober therefore retains the tree's geographic economy and modestly lowers its access peak, while paying a clear load penalty against Clos.

DualDam-4 isolates the canal choice: it keeps the same ponds and selected spine pair but connects each dam directly to both. It uses 0.407 ± 0.004 times Clos's access length, $24.7\pm0.7\%$ more than Bober-4, while its achieved all-link peak is $25.3\pm2.0\%$ lower. Thus neither construction dominates even under the simplified length metric.

Pond size is the tuning knob. At $L=96$, $p=2$ reaches 1.032 ± 0.009 times Clos all-link load with 0.437 ± 0.005 times Clos's modeled access length. This apparent load parity occurs because Clos is core-limited; the access-only sweep in Fig. 3 shows that pond uplinks still concentrate traffic. Increasing p concentrates the union of member-RU flows. The curve is not monotone in cable because Morton grouping, centroids, and duplicate-canal sharing change together: at the mean, $p=6$ uses less cable and load than $p=8$. A biologically larger pond is not automatically an engineering improvement.

Fig. 3 also shows that the access result is structural rather than a 96-RU accident. From 24 to 96 RUs, paired means put Bober-4 6.6–11.5% below the tree's access peak and roughly three times above dual Clos. It has fewer long uplinks, but each shared dam uplink carries several RU unions; scaling the deployment does not make that aggregation penalty vanish.

Late branching is not decorative. In the independently allocated ablation at $L=96$, the late-replication variant's peak is 44.7% below the early-copy overlay and its aggregate core-facing DL load is 67.6% lower. Because the traffic semantics alter selected routes, these are end-to-end design differences rather than a route-held-fixed causal estimate of replication

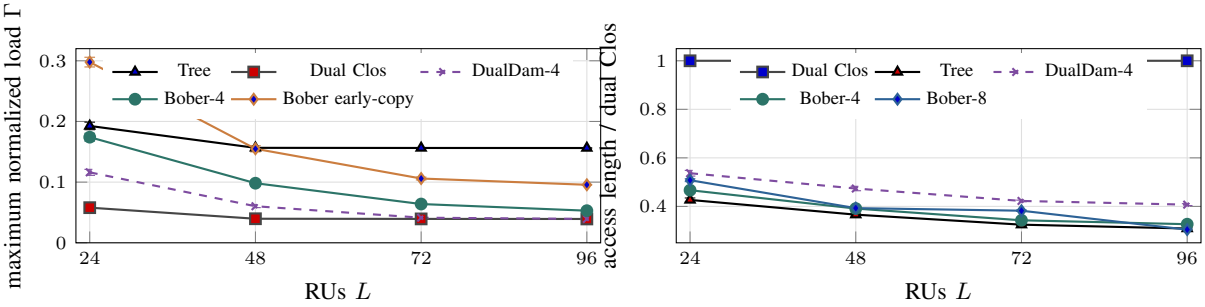


Fig. 2. Spatially uniform scaling with 95% CIs (some smaller than markers). Left: all-link peak; DualDam-4 exposes the canal penalty. Right: normalized modeled access length.

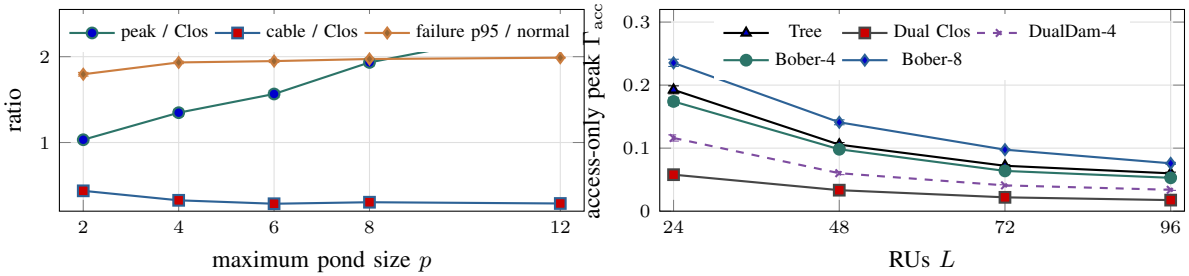


Fig. 3. Sensitivity under spatially uniform placement. Left: pond size trades peak load and protected-failure inflation against modeled length. Right: the access-only view removes the tree's small core cut.

alone. Bober-4's core-facing DL load is also 21.5% below dual Clos, even though its peak access link is higher. The architecture moves the bottleneck outward; it does not erase it.

In Table III, "Core DL sum" is aggregate DU-to-spine load, not a link peak. Pond grouping reduces recipient spines, so Bober-4 uses less core DL work than Clos despite an extra mean access hop. Bober-4 and early-copy share the construction; their hops differ because independent allocation sends the larger early-copy increments over more spillways.

Hotspots are the less flattering test. Table IV uses paired ratios at $L = 96$. Bober-4's all-link peak becomes 2.10 ± 0.05 times Clos, although it remains $34.0 \pm 3.6\%$ below the single-root tree. The absolute mean peak more than doubles from 0.0529 to 0.1113. A planner should shrink or split hot ponds rather than preserve their geometry.

Finally, the 100% primary-uplink reachability entry for Bober is the deterministic consequence of Proposition 1, not an empirical availability estimate. The nontrivial experiment is the load after fixed switching: Bober-4's mean per-trial 95th-percentile failure load is 1.93 times its normal-state maximum, compared with 1.11 for dual Clos and 1.37 for DualDam-4.

TABLE II
SPATIALLY UNIFORM CASE AT $L = 96$ (30-SEED MEANS)

Topology	Γ	Γ_{acc}	Cable/Clos	Reach./fail. p95
Dual Clos	.0393	.0176	1.000	100% / 1.11
DualDam-4	.0393	.0339	.407	100% / 1.37
Bober-4	.0529	.0529	.327	100% / 1.93
Bober-8	.0759	.0759	.304	100% / 1.97
Tree	.1562	.0600	.309	0% / –

TABLE III
MECHANISM INDICATORS, UNIFORM $L = 96$ (30-SEED MEANS)

Topology	Access links	Mean hops	Core DL sum
Dual Clos	192.0	2.00	.621
DualDam-4	144.0	3.00	.589
Tree	120.0	3.00	.376
Bober-4	139.2	3.14	.487
Bober-8	117.6	3.14	.515
Bober early-copy	139.2	3.24	1.503

TABLE IV
BOBER-4 ROBUSTNESS AT $L = 96$ (PAIRED 30-SEED MEANS)

UE layout	Γ	/Clos	/Tree	Acc./Clos	Fail. abs./rel.
Uniform	.0529	1.347	.339	3.036	.102/1.93
Hotspot	.1113	2.096	.660	2.106	.202/1.82

The smaller hotspot ratio in Table IV is not a safety gain: absolute post-failure p95 load nearly doubles from 0.102 to 0.202 because the normal state is already congested. A canal without reserved capacity is merely a line on a diagram.

VI. CAPACITY AND DEPLOYMENT IMPLICATIONS

Let $\mathcal{F}$ be the modeled primary-uplink failures and $L^{(f)}$ the fixed-switching load vector, with $f = \varnothing$ denoting normal operation. The exact robust envelope is

$$\widehat{L}_e = \max_{f \in \mathcal{F} \cup \{\varnothing\}} L_e^{(f)}. \tag{6}$$

A capacity vector accommodates every modeled state iff $C_e \geq \widehat{L}_e$ for every link. The simulator reports scalar ratios $\|L^{(f)}\|_\infty / \|L^{(\varnothing)}\|_\infty$; it does not perform capacity admission.

A production planner should evaluate (6) against measured demands and reserve the result, not multiply every link by the reported p95 ratio.

Four rules follow: begin at $p = 2$ and merge only after normal- and failure-state capacity checks pass; replicate only common pre-RU-processing payload; add local redundancy when drops or canals enter the service objective; and treat canals as targeted, not universal, protection. If Bober-4's 93% mean p95 load inflation is unacceptable, the beaver loses that site; exact reserves must follow (6).

The matched DualDam-4 baseline has $R+2A = 144$ access links and replaces only each canal-plus-neighbor segment with a direct alternate uplink. Under uniform placement its access peak is $35.7 \pm 1.9\%$ below Bober-4's and failure p95 is 1.37 rather than 1.93, at a 24.7% length premium. Under hotspots it lowers achieved peak by only $5.8 \pm 2.4\%$, while relative failure p95 is 1.16 rather than 1.82. Direct dual-homing lowers failure-load inflation in the model, while canals shorten access length; neither is a free resilience mechanism. Since Euclidean length omits trenches, ports, conduits, and common risks, a site-specific cost model could reverse the apparent frontier.

VII. Limits and Conclusion

This is a controlled synthetic graph experiment, not a calibrated digital twin, PHY model, or packet testbed. It omits queue dynamics, propagation calibration, timing asymmetry, switch radix prices, control-plane convergence, DU migration, and radio-quality changes during failures. Normalized lognormal demand is intentionally synthetic, and common-bit replication depends on the stated functional split. Morton ordering is deterministic and cheap but not hotspot-aware; a planner could split ponds using measured union load instead.

The greedy allocator has no optimal benchmark or approximation guarantee, and its decreasing-demand order may hide better joint assignments. Using one heuristic across topologies does not equalize its topology-dependent optimality gap; reported values are achieved loads, not topology-optimal estimates. Failure trials treat listed physical links independently: a shared duct, dam, spine, or power loss can invalidate both installed routes. Confidence intervals capture only seed variation, not these modeling choices. Consequently, the experiment supports comparative topology statements, not latency, schedulability, convergence, or availability guarantees.

Within that scope, Bober-Clos is neither a punchline nor a free lunch. Local aggregation and canals can cut modeled long access-edge length and retain reachability after a primary pond-uplink failure; a Clos core and late replication prevent the design from degenerating into a dam cascade. The price is a pond-size-dependent peak-load factor, nearly twofold mean p95 load inflation under local failover, and unprotected local drops. Here $p = 2$ is the least concentrated ponded point; deployment requires site-specific cost and failure data. This applies only to common downlink bits before RU-specific processing, not lower splits carrying distinct precoded or IQ traffic. Bober at the edge, Clos in the core—and no packets left to hibernate.

Appendix: Agentic AI Disclosure

System, source, and workflow. An OpenAI Codex desktop agent (GPT-5 family; exact checkpoint undisclosed) generated all text, code, experiments, figures, references, and typesetting. The 14-page seed PDF names OpenAI as author; its model and prompt are unknown. Tools were browser research, AI subagents, standard-library simulation, built-in image generation, LaTeX/`IEEEtran`, and rendered-PDF inspection. The human supplied this AI-authored seed and prompts, but no prose or edits.

Initial prompting strategy. The Research Director asked the agent to choose a soon-closing non-flagship venue, ground and format the satirical Bober-Clos concept with literature and simulation, retain Bober and the satire, act as first author, and accept prompts but no manual edits.

Subsequent prompting. Later prompts requested five pages, Bober artwork, AI-review gaming/"exploits," a satirical "MUST ACCEPT" cuteness claim, less identity detail, preflight, and final pass. They supplied goals, not prose, analysis, code, citations, results, or composition. Verbatim prompts and checksums are in the audit bundle.

Constraints and review intervention. The agent kept checksummed controls and limits, added the missing DualDam-4 baseline after hostile review, generated all revisions, and retained unfavorable results. The visible Cuteness Corollary is zero-weight. There is no hidden/clipped text, metadata directive, covert prompt injection, or fabricated evidence. Constraints were five IEEE-style pages including disclosure/references, reproducible synthetic results, valid UL/functional-split semantics, transparent AI-first authorship, and no manual edits. One acceptance cannot establish the need for human review.

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